TriFrame CV: Can Pictures of Stock Charts Predict Risk?

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GitHub repository: github.com/harshatangirala/triframe-cv (includes the full working paper)

Abstract

TriFrame asks a narrow question. Can pictures of a stock's price chart, drawn at weekly, daily and hourly scales, help a deep-learning model forecast the next 20 days of risk better than ordinary numbers computed from the same prices? We built the whole pipeline to find out: 236,520 chart sets from 500 US stocks (2001 to 2022), seven pretrained vision networks used both frozen and fine-tuned, gradient-boosted heads, strict chronological testing and a Streamlit dashboard that reads TradingView screenshots.

Risk turned out to be predictable. On the 2020 to 2022 test years the chosen model explains 50.3% of the variation in 20-day volatility (R^2 0.503) and ranks stocks that later fall 10% above those that do not 65% of the time (AUC 0.648). On squared error it beats the GARCH, EWMA and rolling models, and its edge over HAR is not significant. The chart images did not add a significant gain in overall R^2 on those years, though a later test finds a within-month gain of 0.020 (Section 6). 5-day direction stayed at a coin flip (AUC 0.510), so the system defaults to NO TRADE and is presented as a risk tool.

1. Dataset selection

Our dataset holds daily and regular-session hourly OHLCV bars for 979 US stocks, long enough to span several market regimes. Every run first checks 3,827 files against their manifests.

This run uses 500 of those stocks, drawn at random with a fixed seed. The sample deliberately contains the 60 stocks of our pilot and the 120 of the second run, so all three runs can be compared on identical rows. One example, a *chart set*, is one stock on one trading day: three chart images, 33 numeric features and three labels. We kept 236,520 of them, with decision dates from October 2001 to December 2022.

The three targets cover risk and return:

Target	What it measures	Type
Volatility (V)	log realised volatility of daily returns over the next 20 sessions	regression
Drawdown (D10)	price trades 10% or more below the next day's open within 20 sessions	yes / no
Direction (O5)	5-day return beats the median stock on the same date	yes / no

Time is never shuffled. A row joins a period only if its whole 20-day label window ends inside it, which removed 1,913 boundary rows.

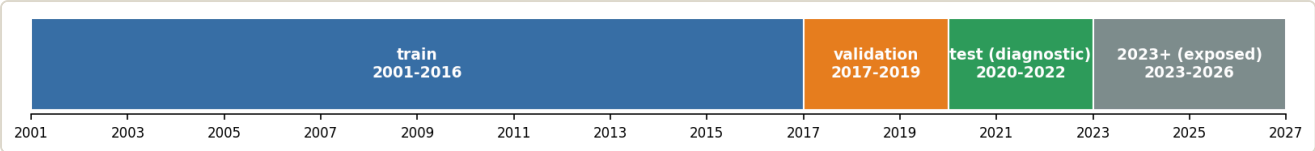


Figure 1. Train, validation and test periods. The 2023+ period is not used.

Period	Years	Chart sets	Stocks	10% drawdown rate
Train	2001-2016	165,421	489	22.7%
Validation	2017-2019	35,145	492	17.8%
Test	2020-2022	35,954	500	30.2%

Two honesty notes. The 2020 to 2022 test years had already been read by our earlier runs, so here they act as a historical check rather than a fresh test. The 2023 onward holdout was opened twice before, so this run uses it for nothing.

2. Pre-processing strategy

The charts are drawn by our own deterministic renderer, not screenshots. Each view is a black-and-white OHLC bar chart, 128 pixels high and 3 pixels per bar, with a 100-pixel price panel above a 26-pixel volume panel. No text, no gridlines, no colour. The weekly view shows 104 bars (about two years), the daily view 126 bars (six months) and the hourly view 140 regular-session bars (about four weeks).

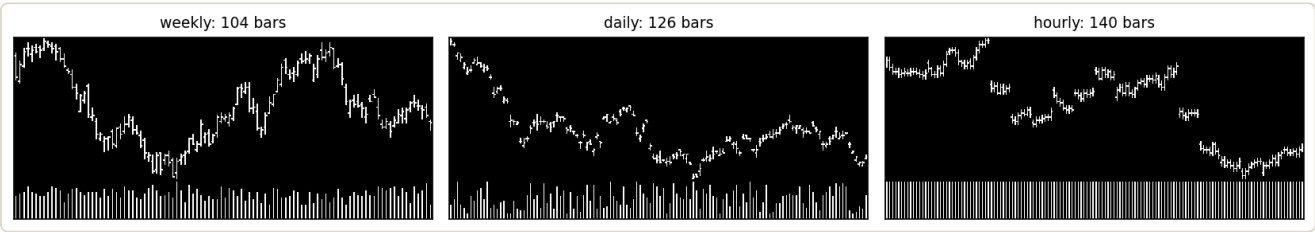


Figure 2. One chart set as the networks see it (drawn here from synthetic prices).

Prices are put on a log scale and stretched to fill each panel. That choice matters. A picture shows the *shape* of recent moves but never their size, so the images cannot see the volatility level directly. The numbers carry it. Before entering a network, each 0/1 image is copied into three channels, resized to 224 × 224 with antialiasing and normalised with that network's own ImageNet mean and standard deviation.

The 33 numeric features come from the last 126 daily bars, the same window a daily screenshot shows. They cover recent returns split into overnight and intraday parts, volume surprises, mean-reversion signals such as short RSI and z-scores, volatility estimators (close-to-close, Parkinson and Garman-Klass over 1, 5 and 22 days, plus EWMA) and amplitude measures such as ATR and the largest drawdown in the window.

Every fitted transformation learns from the training years only. Missing numbers are filled with training medians and then standardised with training means and deviations. Each network turns each view into an embedding of 384 to 1,280 values, which PCA compresses to 128 per view; the PCA keeps between 92% and 98% of the variance. Image features are standardised and clipped at ±10 standard deviations.

The dashboard runs this exact code, and the saved bundle reproduced the notebook's test predictions to within 4×10^{-5}.

3. Model selection and configuration

We picked seven pretrained networks from the timm library to span the main design families: convolutional and transformer, 4 to 86 million parameters, supervised and self-supervised pretraining. All are light enough to run on a CPU inside the dashboard.

Network	Family	Pretrained on	Parameters (M)
ViT-B/16	Transformer	ImageNet-21k, then 1k	85.8
ConvNeXt-Tiny	CNN	ImageNet-12k, then 1k	27.8
Swin-T	Transformer	ImageNet-1k	27.5
DINOv2 ViT-S/14	Transformer	LVD-142M (self-supervised)	21.6
ResNet-18	CNN	ImageNet-1k	11.2
MobileNetV3-Large	CNN	ImageNet-1k	4.2
EfficientNet-B0	CNN	ImageNet-1k	4.0

Three ResNet-18s with random weights serve as controls. If they matched the pretrained networks, pretraining would be doing nothing.

On top of the compressed embeddings sit simple heads, one per target: ridge regression for volatility, logistic regression for the yes/no targets and XGBoost trees (depth 3 to 7, learning rate 0.05 or 0.10, early stopping up to 2,000 trees). Each head was tried on four input sets: images only, numbers plus all three views, numbers plus one view, and numbers only. Classical volatility models (HAR, EWMA, a rolling 22-day estimate and, in a later local run, GARCH(1,1)) set the floor. Probabilities are calibrated with isotonic regression on the validation years.

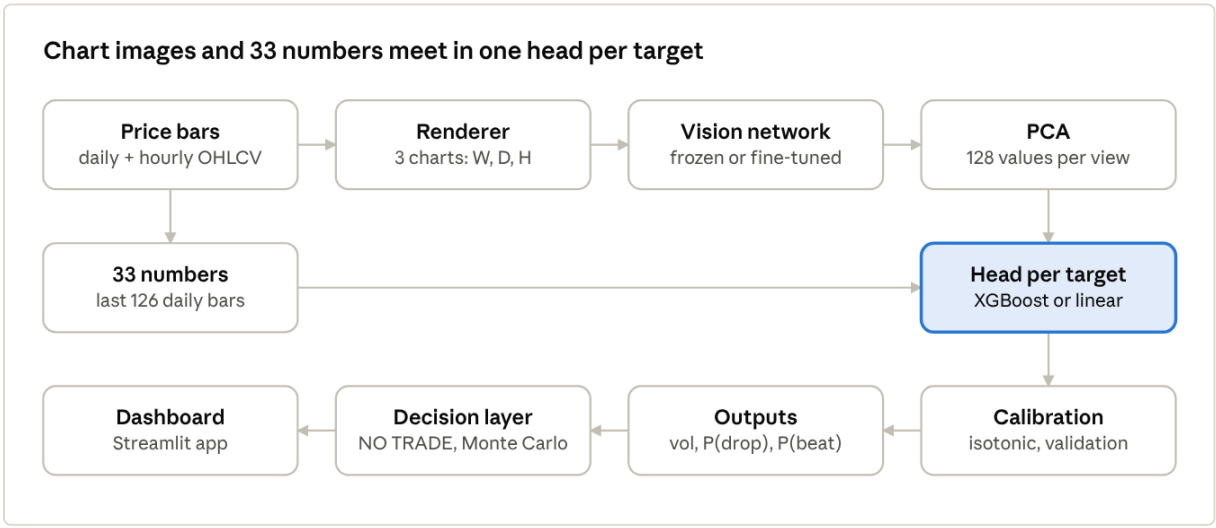


Figure 3. TriFrame architecture, from price bars to the dashboard

The selection rule was written down before testing. The best validation score wins, but a model that uses images wins only if its gain over numbers alone has a 95% confidence interval above zero and holds in every validation year. For volatility, numbers plus the three views from a fine-tuned ConvNeXt-Tiny passed easily: +0.046 R^2 (CI +0.035 to +0.059), positive in 2017, 2018 and 2019. The same features won for drawdown. Direction kept numbers only.

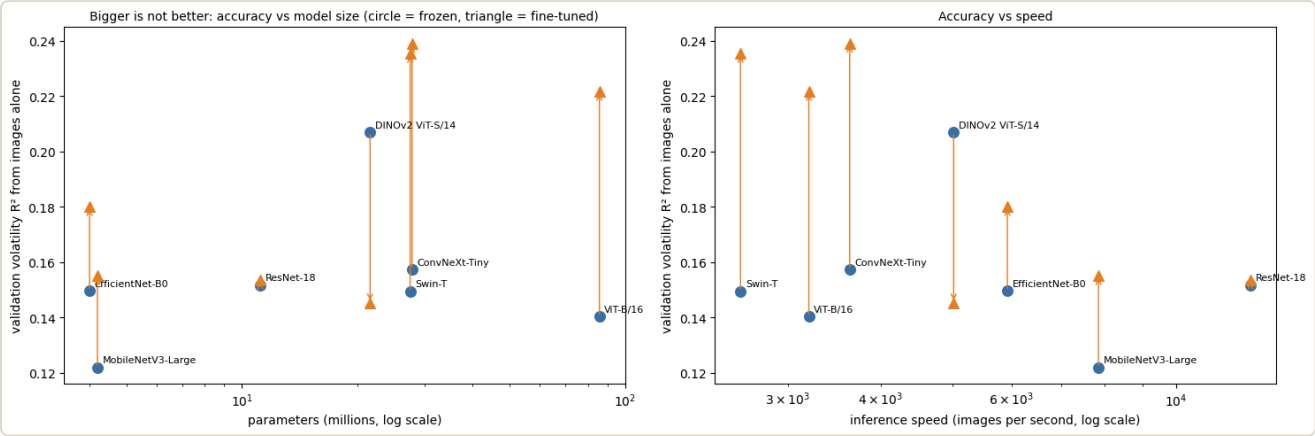


Figure 4. Bigger is not better: validation R^2 from images alone against parameters and speed (circles frozen, triangles fine-tuned).

A decision layer turns outputs into actions. A stock is flagged high-risk when predicted volatility reaches 33.6% a year or drawdown probability reaches 30.3%, the top fifth of validation readings. A direction call would appear only if the direction model passed both its validation and test checks.

4. Training and fine-tuning process

We tried 1,881 model configurations and logged every one. All choices were made on the 2017 to 2019 validation years, with confidence intervals from 1,000 month-block bootstrap resamples, because neighbouring stock-days share overlapping 20-day windows and are not independent.

Fine-tuning trains each network end to end. One backbone reads all three views, a small head adds the 33 numbers, and the loss combines mean squared error on volatility with binary cross-entropy for the two yes/no targets. We used AdamW (learning rate 3×10^{-5} for the backbone, 10^{-3} for the head, weight decay 0.05), batch size 64, a short warm-up followed by cosine decay, gradient clipping at 1.0 and bfloat16 mixed precision on an A100. There was no augmentation. A flipped chart tells a different market story.

Each epoch drew a fresh 60,000 of the 141,525 chart sets whose labels end before 2015. Early stopping watched 15,000 rows from 2015 and 2016, inside the training years, with patience 2 and at most 8 epochs. The validation years were never touched, a fix over our earlier run, which had early-stopped on them. A controlled study then trained only the last 25% or 50% of the two best networks, tried a risk-only loss and repeated the best setup with two extra seeds. In all: 15 runs, 158 GPU-minutes.

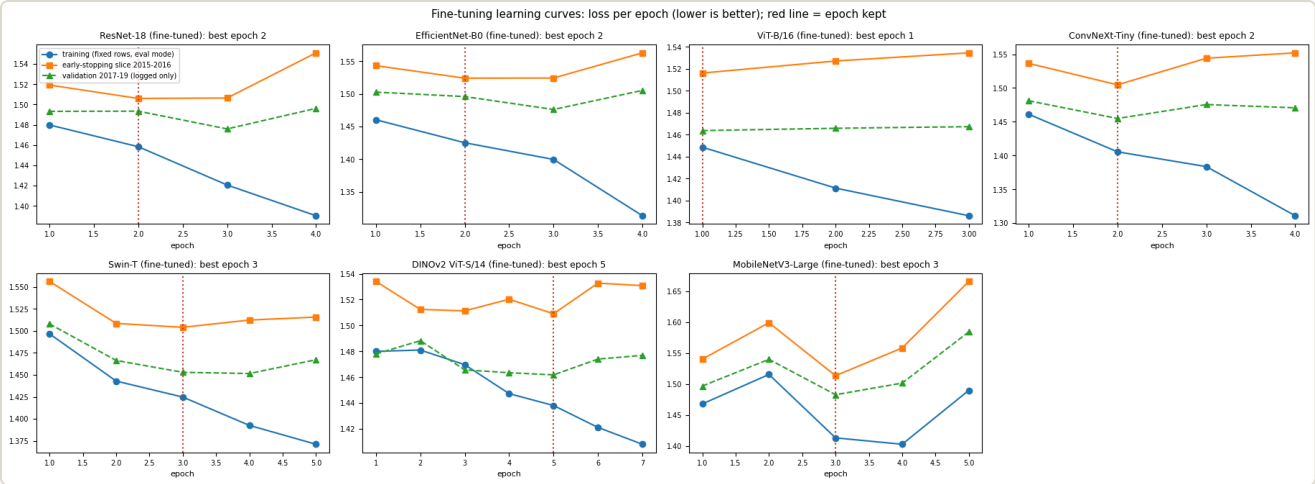


Figure 5. Loss per epoch; the red line marks the epoch kept.

Fine-tuning raised validation volatility R^2 in 9 of 9 comparisons, significantly in 7. ConvNeXt-Tiny gained most (0.561 to 0.592), followed by ViT-B/16 (+0.028) and Swin-T (+0.025). Training only the last quarter or half gained almost nothing (+0.001 and +0.002). The low-level filters, tuned on photographs, are what needed to adapt.

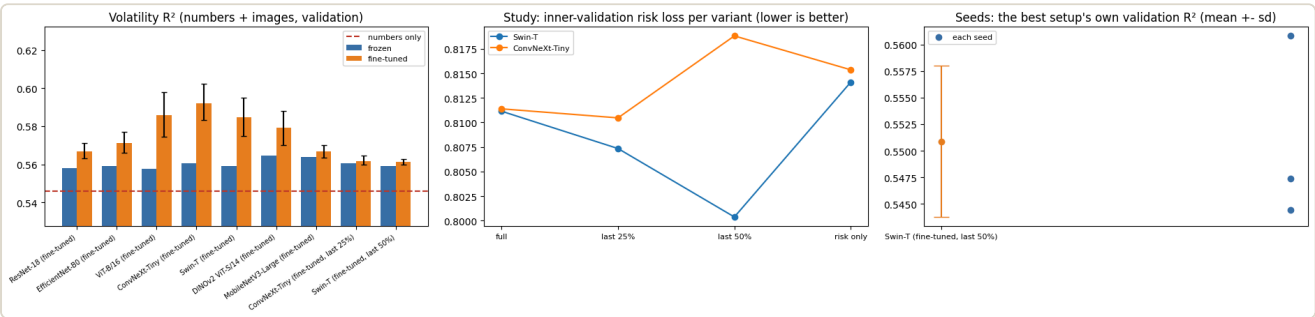


Figure 6. Frozen against fine-tuned R^2 with 95% CIs, the study variants and the seed spread.

Did training reach the global minimum? For the convex heads, provably yes: the ridge gradient at its closed-form solution is 6.1×10^{-16}, and five random starts of both ridge and logistic regression end at the same answer. Boosted trees keep the lowest point of their validation curve. Neural networks are not convex, so no method can certify their global minimum. We can show that the kept checkpoint has the lowest held-out loss and that, on random planes through the weights, the trained ConvNeXt sits at the bottom of its valley.

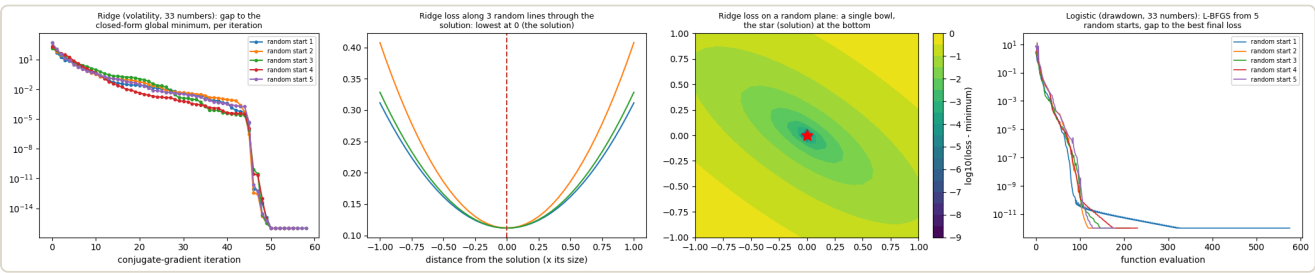


Figure 7. Convex heads: five random starts, one minimum.

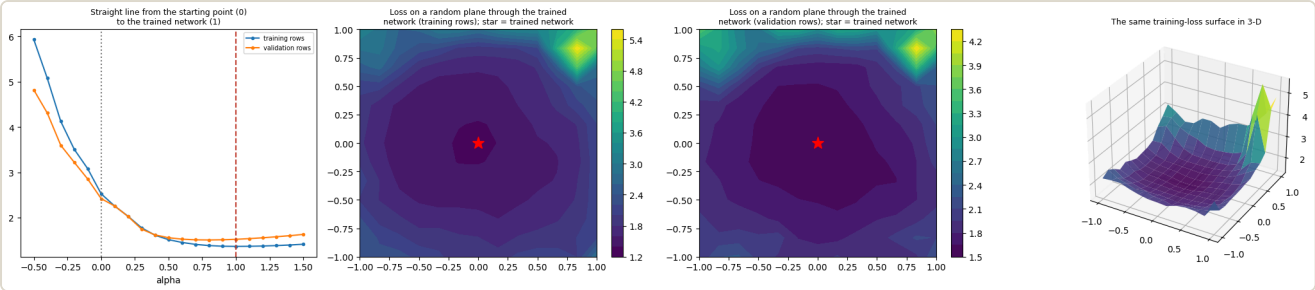


Figure 8. ConvNeXt-Tiny loss landscape (star = trained network).



Figure 9. Boosting curves; the dashed line marks the kept minimum.

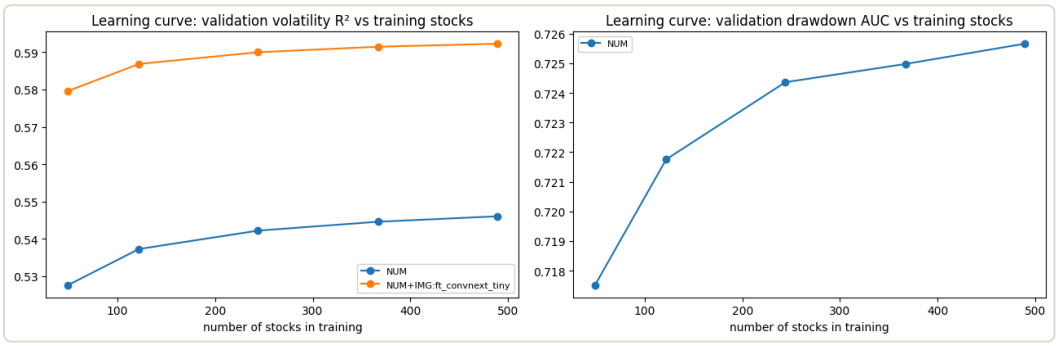


Figure 10. More stocks help, with diminishing returns after about 250.

5. Evaluation metrics and results

Volatility is scored with R² and RMSE on log volatility. The yes/no targets get ROC-AUC, plus precision, recall and F1 at a threshold fixed on validation (0.223 for drawdown, 0.439 for direction).

Target	Metric	Validation	Test (95% CI)	Numbers only, test
Volatility	R²	0.592	0.503 (0.25 to 0.65)	0.500
Volatility	RMSE	0.321	0.381	0.382
Drawdown	ROC-AUC	0.740	0.648 (0.59 to 0.69)	0.653
Drawdown	Precision / recall / F1	0.36 / 0.55 / 0.43	0.39 / 0.65 / 0.49	0.37 / 0.74 / 0.49
Direction	ROC-AUC	0.516	0.510 (0.500 to 0.521)	0.510

On the test years the volatility model reaches R² 0.503, against HAR 0.478, EWMA 0.448, GARCH 0.433 and a rolling 22-day estimate 0.416. The gain over GARCH is significant; the gain over HAR is not. Under QLIKE, which punishes under-forecasts, the simple models win. Rank correlation with realised volatility is 0.74.

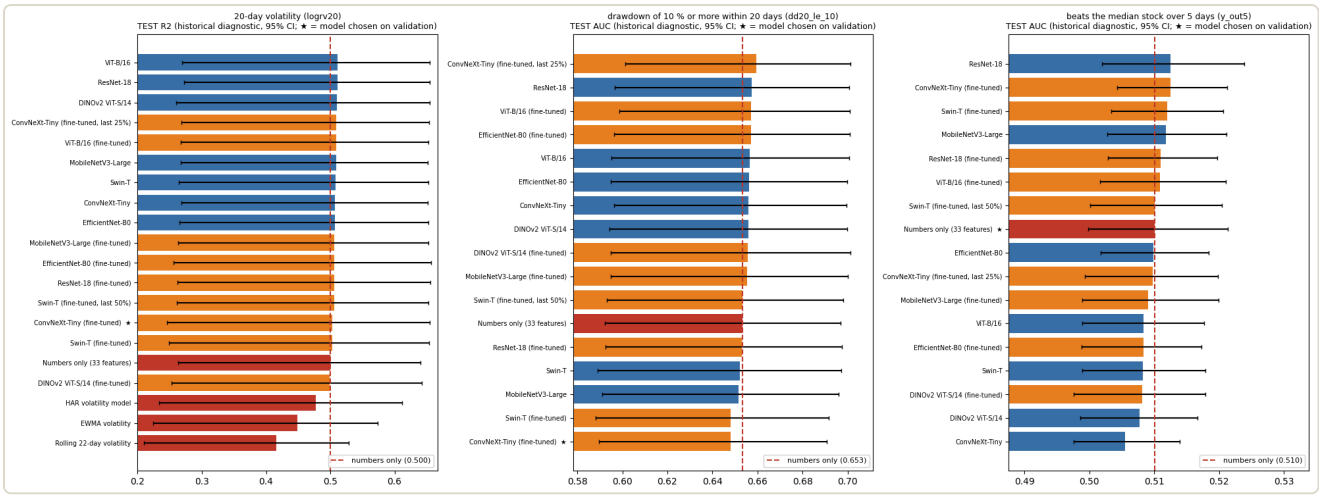


Figure 11. Every candidate on the test years with 95% CIs (star = the model chosen on validation).

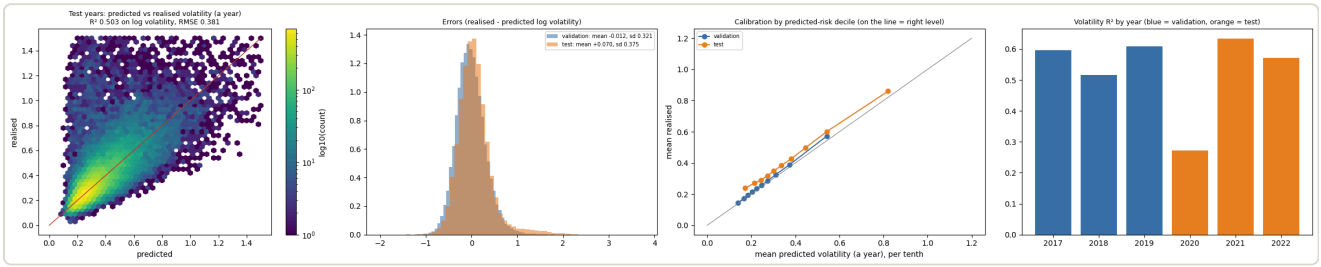


Figure 12. Volatility: predicted against realised, errors, calibration by decile and R² by year.

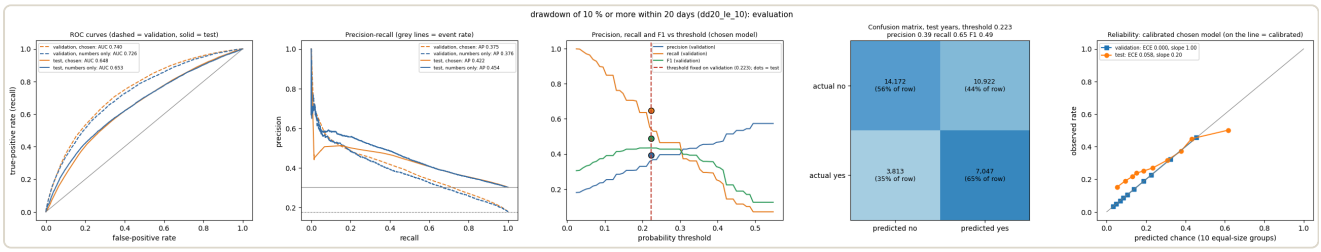


Figure 13. Drawdown: ROC, precision-recall, thresholds, confusion matrix and reliability.

The decision layer works as intended. On the test years it flagged 48.3% of cases, and those cases really were riskier: realised volatility of 55.7% a year against 29.8%, and 10% drawdowns 40.1% of the time against 21.0%. The Monte Carlo simulator's 90% band held 88% of real 20-day outcomes.

Scaling up paid off for volatility. On exactly the same rows, the 500-stock model raised test R² over the 60-stock pilot from 0.496 to 0.510 and over the 120-stock run from 0.503 to 0.513.

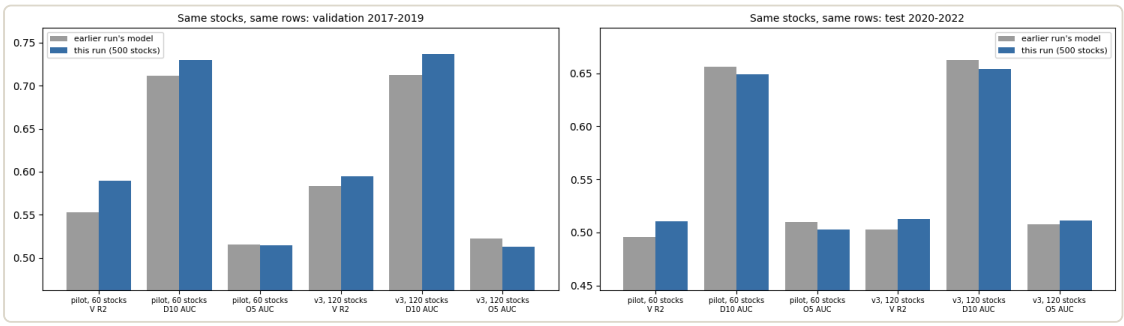


Figure 14. Earlier runs against this run on the earlier runs' own stocks and rows.

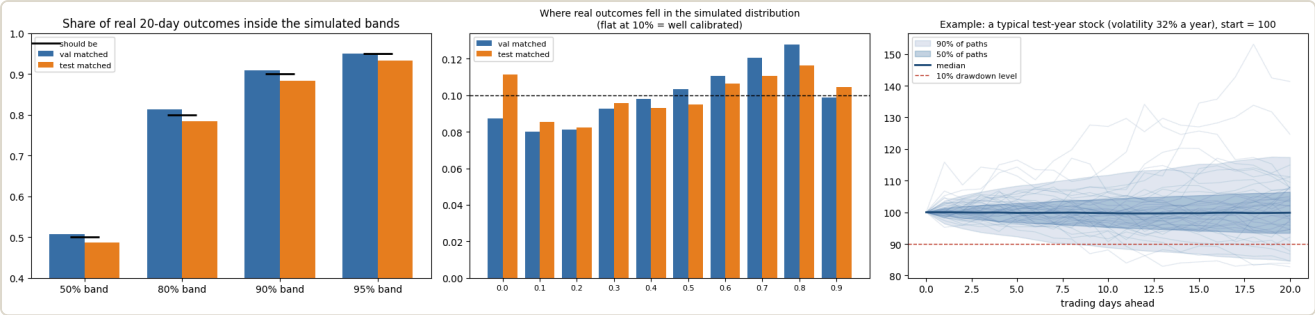


Figure 15. Monte Carlo band coverage and an example fan of paths.

6. Analysis of performance and potential biases

Chart images help on validation (+0.046 R^2) but not on the test years, where adding them changed volatility R^2 by +0.003 (95% CI −0.018 to +0.026) and direction AUC by −0.002. The pictures do carry real information: the best network reaches R^2 0.207 from images alone, against about −0.02 for random-weight networks and −0.123 with shuffled labels, which also rules out leakage. A later analysis splits the error. Within each month, images raise R^2 by +0.020 (95% CI 0.011 to 0.027), while the market-wide level forecast does not improve, so the two offset in overall R^2. Hourly realised volatility does not explain the gain. We chose this test after seeing the data, so it needs confirming on new data. Because the renderer stretches every chart to fill its panel, the images cannot see the volatility level at all.

One year explains most of the drop from validation to test. Volatility R^2 was 0.27 in 2020, against 0.63 in 2021 and 0.57 in 2022. The COVID crash was a market-wide shock that no single stock's history foretold, and our 2017 to 2019 validation years held no crisis to prepare the models for one.

Four biases deserve naming:

- **Survivorship.** The universe contains only stocks that still trade, so risk is probably understated for firms that later failed.
- **Base-rate shift.** The drawdown rate rose from 17.8% in validation to 30.2% in test, and calibration error rose to 0.058.
- **Selection luck.** Choosing among many candidates on one three-year window rewards noise. The chosen drawdown model ranked last of 17 on test, though all 17 sit within 0.012 AUC.
- **Reused test years.** Earlier runs had read 2020 to 2022, so they now serve as a historical check. A clean test needs data after October 2026.

One metric needs care. Direction's F1 of 0.67 at recall 1.00 is what flagging every stock scores, not skill.

7. Ethical and societal implications

The biggest risk is misuse. A dashboard that turns a screenshot into a number invites people to trade on it, and retail investors tend to trust confident software. We designed against that. The default decision is NO TRADE, direction calls stay hidden because the model has no edge, every output carries a plain-language explanation and a risk tier, and the dashboard states that it is research output, not investment advice.

Model errors are not spread evenly. They peak in crises, the moment users most need accurate risk estimates, and survivorship bias may make fragile companies look safer than they are. Calibrating probabilities and publishing the 2020 weakness are partial answers. Neither replaces human judgement.

Transparency was a deliberate choice. Code, reports, executed notebooks and the frozen model choices are public, and so is the negative result on images. Claims that AI "reads charts" spread easily, while null results rarely get published. Market data and trained weights stay private because the data licence has not been cleared for redistribution.

Compute has a cost too: fine-tuning alone took 205 minutes on an A100. Cached embeddings, PCA compression and early stopping kept the whole run to about four and a half hours. The models use only market data, and screenshots uploaded to the dashboard are read in memory, never stored.

Disclosure of AI use. AI assistants (principally Anthropic's Claude Code, with Google's Antigravity and OpenAI's ChatGPT for earlier work and review) wrote most of the code, proposed much of the statistical design, ran analyses and drafted this report under the author's direction. The author takes responsibility for the content. Appendix B of the working paper itemizes this use.